COURSE 4023 - REVISION 2021

Automobile Engineering

A complete diploma engineering handbook for automobile structure, IC engine systems, transmission, steering, suspension, braking, electric vehicles, hybrid vehicles, emission control and Motor Vehicle Act review.

PROGRAM
Diploma in Mechanical Engineering

SEMESTER
4

CREDITS
4

PERIODS
60

> □ handbook automobile subject-□□ system flow □□□ □□□□□□□□□□□□□□□: engine power generate □□□□□□□□□□, transmission torque wheels-□□□□□□ □□□□□□□□□□□□□, steering/suspension/brakes vehicle control □□□□□□□□□□, EV systems battery + motor + power electronics □□□□□□□□□□□□□□.

Course Overview and Redesigned Syllabus Roadmap

Official information extracted from the supplied syllabus. Missing official information is marked instead of invented.

Objectives

- Acquire knowledge about automobile classification, structure, IC engine, cooling, lubrication, fuel, ignition and governing systems.
- Understand the transmission system in automobiles.
- Familiarize suspension, steering and braking systems.
- Identify electric, hybrid-electric and plug-in hybrid vehicles.
- Familiarize with Indian Motor Vehicle Act.

How to study

1. Learn power flow from engine or motor to wheels.
2. For every system, learn need, parts, working, diagram and faults.
3. Use tables for comparisons.
4. Practice diagrams repeatedly.

Confirmed official information

Item	Information
Course code	4023
Course title	Automobile Engineering
Program	Diploma in Mechanical Engineering
Semester	4
Credits	4
Category	Program Core
Periods	4 per week, 60 per semester
Practicals / experiments	Not specified in supplied syllabus/source.
Official final model paper pattern	Not specified in supplied syllabus/source. Practice model paper is created for revision only.

Course outcomes

CO	Outcome	Hours	Level
CO1	Describe classification, structure, engine components, cooling, lubrication, fuel, ignition and governing systems.	18	Understanding
CO2	Explain the transmission system in automobiles.	17	Understanding
CO3	Explain ignition, suspension, steering and braking systems.	15	Understanding
CO4	Compare EV, HEV, PHEV, emission control and Motor Vehicle Act.	10	Understanding

Roadmap

Module	CO	Hours	Chapter	Exam focus
1	CO1	18	Automobile structure, IC engine and engine systems	Parts, block diagrams, cooling/lubrication/fuel/ignition working
2	CO2	17	Transmission system	Clutch, gearbox, fluid coupling, torque converter, differential and axles
3	CO3	15	Suspension, steering, wheels, tyres and brakes	Figures, definitions, hydraulic braking, ABS and EBD
4	CO4	10	EV, HEV, PHEV, emission and Act	Comparisons, power flow, battery, regeneration, safety

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CO1 - 18 HOURS

Module 1: Automobile Structure, IC Engine and Engine Systems

Classification, basic structure, engine components, cooling, lubrication, fuel, ignition, battery, charging, starting and governing systems.

Official module outline

Point	Description	Hours

M1.01	Classification and basic structure of automobile; major IC engine parts	5
M1.02	Cooling systems	1
M1.03	Water cooling components	2
M1.04	Lubrication systems	1
M1.05	Petrol and diesel fuel systems	3
M1.06	Ignition systems	1
M1.07	Lead acid battery, charging and starting	3
M1.08	Governing system and types	2

Malayalam support

> Fuel system mixture/pressure □□□□□□□□□□□□□, ignition spark □□□□□□□□□□□□□□, cooling heat control □□□□□□□□□□□, lubrication friction □□□□□□□□□□□□□□□. □□ □□□□□□□□□□ engine smooth running □□□□□□□□□□.

Common mistakes

- Writing only engine parts and ignoring chassis.
- Confusing wet liner and dry liner.
- Saying radiator cools engine directly; coolant carries heat.
- Mixing petrol carburetor and diesel injector working.

Textbook chapter explanation

Classification and structure

Automobiles are classified by purpose, load, fuel, number of wheels, drive layout, engine position and transmission type. A vehicle includes chassis, frame/body, power plant, transmission, suspension, steering, brakes, wheels and electrical systems.

Engine components

Cylinder block holds cylinders and crankcase. Cylinder head closes the cylinder. Gaskets seal joints. Liners provide wear surface. Piston, rings and pin transfer combustion force to connecting rod and crankshaft. Camshaft operates valves. Flywheel smoothens speed.

Cooling

Cooling removes excess heat. Air cooling uses fins and air flow. Liquid cooling circulates coolant through water jackets, pump, radiator and thermostat. Thermo siphon uses density difference; pump circulation uses forced flow.

Lubrication

Lubrication reduces friction, wear, heat, noise and corrosion. Splash, forced and mist/petroil lubrication are syllabus types. Forced lubrication is common in multi-cylinder engines.

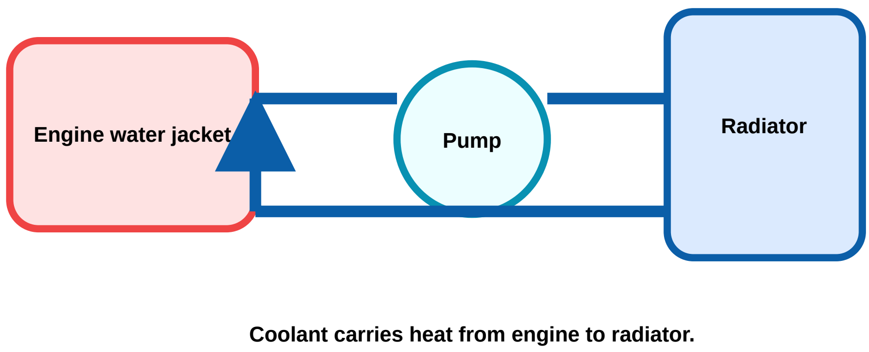
Fuel and ignition

Petrol engines need correct air-fuel mixture through carburetor, SPFI or MPFI. Diesel engines require high-pressure injection through pump and injector. Battery/electronic ignition provides timed spark in SI engines.

Battery, charging, starting, governor

Lead acid battery supplies starting and ignition power. Alternator charges battery. Starter motor cranks engine. Governor controls engine speed by quantity, quality or hit-and-miss methods.

Diagram lab



Water cooling system

CO2 - 17 HOURS

Module 2: Transmission System in Automobiles

Clutch, gearbox, fluid coupling, torque converter, overdrive, propeller shaft, universal joint, final drive, differential and live rear axle.

Textbook explanation

Need of transmission

Engine torque is useful only in a limited speed range. Transmission multiplies torque during starting/climbing and gives higher speed during cruising.

Clutch

Clutch connects and disconnects engine from gearbox, permits smooth starting and gear shifting. Single plate, multi plate, diaphragm and centrifugal clutch must be compared.

Gearbox

Gearbox provides speed ratios, neutral and reverse. Sliding mesh is simple but noisy. Constant mesh keeps gears engaged. Synchromesh equalizes speeds before engagement. Epicyclic gears use sun, planet, ring and carrier.

Fluid coupling and torque converter

Fluid coupling transfers torque through fluid and gives smooth drive. Torque converter adds a stator and can multiply torque during starting.

Differential and axle

Differential allows inner and outer wheels to rotate at different speeds during turning. Live rear axle may be semi-floating, three-quarter floating or full-floating.

Formula

Gear ratio

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Gear ratio = teeth on driven gear / teeth on driver gear
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Used to find torque multiplication and speed reduction.

Output rpm

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Output rpm = Input rpm / gear ratio
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If input is 2400 rpm and ratio is 3:1, output is 800 rpm.

Solved example

Gearbox speed

Given: input speed = 2400 rpm, gear ratio = 3:1.

Formula: output rpm = input rpm / gear ratio.

Calculation: 2400 / 3 = 800 rpm.

Answer: Output speed = 800 rpm.

CO3 - 15 HOURS

Module 3: Suspension, Steering, Wheels, Tyres and Brakes

Suspension systems, steering gears, wheel alignment, tyres, braking systems, ABS and EBD.

Suspension

Suspension supports vehicle weight, absorbs road shocks, maintains tyre contact and improves ride comfort. Leaf spring, coil spring, spring shackle and air suspension are important.

Steering

Steering system includes wheel, column, steering gear and linkages. Worm and sector, rack and pinion, recirculating ball and power steering should be explained with diagrams.

Wheel alignment

Camber is wheel tilt from front view. Caster is steering axis tilt from side view. King pin inclination helps return. Toe-in/toe-out affects tyre wear and straight running.

Brakes

Mechanical, hydraulic and air brakes convert kinetic energy to heat. Hydraulic brakes use master cylinder and brake fluid. Disc brakes use caliper and pads. ABS prevents wheel lock; EBD distributes brake force.

Formula

Braking distance estimate

$$s = v^2 / (2 \mu g)$$

For v = 20 m/s and μ = 0.5, s = 40.8 m.

Malayalam support

Suspension shock absorb □□□□□□□□□□, steering direction control □□□□□□□□□□, brakes stopping control □□□□□□□□□□. Wheel alignment □□□□□□□□□ tyre wear, pulling, poor braking □□□□□□□□.

CO4 - 10 HOURS

Module 4: EV, HEV, PHEV, Emission and Motor Vehicle Act

Electric vehicles, batteries, fuel cells, flywheel storage, regenerative braking, hybrids, BS IV/VI and Indian Motor Vehicle Act.

EV architecture

EV uses battery pack, BMS, inverter/motor controller, electric motor, charger, DC-DC converter and control electronics. High-voltage safety is compulsory.

Batteries

Li-ion and NiMH are syllabus types. Parameters include capacity, voltage, energy density, power density, internal resistance, cycle life and temperature range.

Fuel cell and flywheel

Fuel cell converts chemical energy directly to electrical energy. Flywheel stores kinetic energy in a rotating mass and releases it quickly.

Regenerative braking

During braking, traction motor works as generator and returns part of kinetic energy to battery. It improves efficiency but cannot replace friction brakes completely.

HEV and PHEV

Series hybrid uses engine generator and motor drive. Parallel hybrid allows both engine and motor drive. PHEV has larger battery and external charging.

Emission and Act

BS VI is stricter than BS IV. Motor Vehicle Act covers registration, licensing, insurance, permits, offences, penalties, vehicle fitness and road safety.

Formula

Battery energy

$$\text{Energy (Wh)} = \text{Voltage (V)} \times \text{Capacity (Ah)}$$

350 V and 60 Ah gives 21000 Wh = 21 kWh.

Safety

- Use insulated tools and lockout/tagout for EV high voltage work.
- Do not open battery packs without training.
- Verify zero voltage before touching power cables.

Formula / Rule Bank

For non-numerical systems, the rule is the diagram checklist and troubleshooting sequence.

Engine displacement

$$V = (\pi/4) \times D^2 \times L \times n$$

D = bore, L = stroke, n = cylinders. Used to compare engine capacity.

Compression ratio

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CR = (swept volume + clearance volume) / clearance volume
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Higher compression generally improves efficiency but may increase knock tendency.

Gear ratio

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Driven teeth / Driver teeth
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Used in gearbox speed and torque calculations.

Braking distance

$$s = v^2 / (2 \mu g)$$

Shows why high speed sharply increases stopping distance.

Battery energy

$$Wh = V \times Ah$$

Used for EV battery energy estimation.

Solved Examples / Problem Lab

Engine capacity

4-cylinder engine: bore 80 mm, stroke 90 mm.

$V = (\pi/4) \times 0.08^2 \times 0.09 \times 4 = 0.00181\ m^3 = 1810$ cc.

Answer: about 1810 cc.

Gearbox speed

Input = 2400 rpm, ratio = 3:1.

Output rpm = 2400 / 3 = 800 rpm.

Answer: 800 rpm.

Braking distance

$v = 20$ m/s, $\mu = 0.5$.

$s = 400 / (2 \times 0.5 \times 9.81) = 40.8$ m.

Answer: 40.8 m.

Battery energy

Battery = 350 V, 60 Ah.

Energy = 350 × 60 = 21000 Wh = 21 kWh.

Answer: 21 kWh.

Practice and Expected Questions

Objective questions

1. Function of thermostat?

A. Charge battery B. Control coolant flow C. Change gear ratio D. Increase fuel pressure

Answer: B

2. Device allowing different wheel speeds while cornering?

A. Clutch B. Differential C. Radiator D. Governor

Answer: B

3. ABS mainly prevents?

A. Engine knock B. Wheel lock C. Battery sulphation D. Coolant boiling

Answer: B

4. PHEV differs because it?

A. Has no battery B. Can charge from external supply C. Has no engine D. Uses carburetor only

Answer: B

Module-wise expected questions

1. Explain water cooling system.
2. Compare air cooling and liquid cooling.
3. Explain lead acid battery.
4. Explain single plate clutch.
5. Compare sliding mesh, constant mesh and synchromesh gearboxes.
6. Explain differential.
7. Define camber, caster, KPI and toe.
8. Explain hydraulic brake, ABS and EBD.
9. Compare EV, HEV and PHEV.

Industrial troubleshooting

- Overheating: check coolant, fan, thermostat, radiator and pump.
- No-start: check battery, starter, solenoid, ignition and fuel supply.
- Gear shifting fault: check clutch free play, worn plate and synchromesh.
- Brake fault: check leakage, air, master cylinder, pads and ABS sensors.

Fresh Model Question Paper

Practice paper prepared from the syllabus. Not copied from any official question paper.

Course: 4023 Automobile Engineering
Time: 3 Hours
Maximum Marks: 100

Part A - Short answer
1. Define automobile chassis.
2. State three functions of lubrication system.
3. What is a thermostat?
4. Define gear ratio.
5. State the purpose of universal joint.
6. Define camber and caster.
7. What is ABS?
8. List two EV propulsion components.
9. State two areas covered by Motor Vehicle Act.
10. What is regenerative braking?

Part B - Answer any seven
11. Explain major parts of IC engine.
12. Explain water cooling system.
13. Explain lead acid battery.
14. Explain single plate clutch.
15. Compare gearbox types.
16. Explain differential.
17. Explain rack and pinion steering.
18. Explain hydraulic braking system.
19. Compare EV, HEV and PHEV.
20. Explain fuel cell and regenerative braking.

Part C - Long answers
21. Explain cooling, lubrication and fuel systems.
22. Explain complete transmission from clutch to axle.
23. Explain suspension, steering, tyres and brakes.
24. Explain EV architecture, batteries, emission standards and safety.

Complete Model Answers / Scoring Guide

Engine systems

Include need, labelled diagram, components and working. For cooling show radiator, fan, pump, thermostat and coolant flow. For fuel system compare petrol mixture preparation and diesel high pressure injection.

> Avoid writing paragraphs without diagrams.

Transmission

Start with need of torque multiplication. Explain clutch, gearbox, propeller shaft, universal joint, final drive, differential and axle. Use arrows to show torque flow.

Steering and brakes

Define camber, caster, KPI and toe. Draw Ackerman mechanism. For brakes show master cylinder, fluid line, wheel cylinder/caliper and disc/drum. Explain ABS and EBD.

EV and Act

Draw EV power flow: battery, BMS, inverter, motor and wheels. Compare EV, HEV and PHEV. Mention BS VI, high-voltage safety and Motor Vehicle Act areas.

Official references

- Automobile Engineering Vol. I & II - Kirpal Singh
- Automobile Engineering - K. Ramalingam
- Automobile Engineering - R. K. Rajput
- Hybrid, Electric and Fuel-Cell Vehicles - Jack Erjavec & Jeff Arias
- Electric & Hybrid Vehicles - Design Fundamentals - Iqbal Hussain

Online resources from syllabus

1. Springer automotive journal
2. IJAT
3. NPTEL automobile course
4. AFDC electric batteries
5. Motor Vehicles Act official links